
Road Damage Detection

Using YOLOv8 Deep Learning

Technical Report

AI-Powered Infrastructure Analysis System
Documentation & Final Report

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Project Title	Road Damage Detection Using YOLOv8
Dataset	RDD2022 (Road Damage Dataset 2022)
Model	YOLOv8 (You Only Look Once v8)
Framework	Python, Flask, Ultralytics
Report Date	May 2026

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Abstract

Road infrastructure degradation poses a significant challenge for urban maintenance authorities worldwide. Manual inspection of road surfaces is time-consuming, inconsistent, and expensive. This report presents an automated road damage detection system built on the YOLOv8 object detection architecture, trained on the RDD2022 benchmark dataset comprising over 47,000 annotated road images from six countries.

The system detects four categories of road damage: longitudinal cracks (D00), transverse cracks (D10), alligator cracks (D20), and potholes (D40). The trained model is deployed as a RESTful API backend using Flask and integrated with a full web-based user interface, allowing real-time road damage detection from uploaded photographs. The system achieves a target mean Average Precision (mAP@0.5) exceeding 0.60, demonstrating practical utility for infrastructure monitoring applications.

Keywords: Road damage detection, YOLOv8, object detection, deep learning, computer vision, RDD2022, Flask, infrastructure monitoring.

1 Introduction

1.1 Background and Motivation

Road networks form the backbone of urban and inter-city transportation. Deteriorating road surfaces — manifesting as cracks, potholes, and surface deformation — represent not only a public safety hazard but also a significant economic burden. Traditional road inspection methods rely on physical surveys conducted by trained inspectors, which are inherently slow, subjective, and cannot scale to the size of modern road networks.

The emergence of computer vision and deep learning provides a compelling alternative: automated visual inspection systems capable of detecting road damage with speed and consistency far exceeding human inspection. This project leverages the YOLOv8 architecture, the current state-of-the-art single-stage object detector, to build such a system.

1.2 Problem Statement

The core problem addressed by this project is the automatic identification and classification of road surface damage from photographic images. Specifically, the system must:

- Accurately detect the presence of road damage in an uploaded image.
- Classify the type of damage into one of four standard categories.
- Localise the damage by drawing bounding boxes around affected regions.
- Provide a confidence score and severity assessment for each detection.
- Deliver results through an accessible, real-time web interface.

1.3 Project Objectives

1. Design and implement a YOLOv8-based object detection pipeline for road damage.
2. Prepare and preprocess the RDD2022 dataset for YOLOv8 training.
3. Train, validate, and evaluate the model to achieve $\text{mAP}@0.5 \geq 0.60$.
4. Deploy the model via a Flask backend server on Port 5001.
5. Build a production-quality web frontend for end-user interaction.

2 Related Work

Road damage detection has been an active research area since the availability of large annotated datasets. Early approaches used hand-crafted image features (e.g., Gabor filters, edge detectors) followed by classical machine learning classifiers such as Support Vector Machines (SVMs). These methods were sensitive to lighting variations and required significant feature engineering.

With the rise of deep learning, convolutional neural network (CNN)-based approaches became dominant. Region-based CNNs (R-CNN, Faster R-CNN) demonstrated strong detection accuracy but suffered from slow inference speeds unsuitable for real-time applications.

The YOLO (You Only Look Once) family of models introduced single-pass detection, dramatically improving inference speed. The RDD benchmark series (RDD2018, RDD2020, RDD2022) standardised evaluation protocols and provided large-scale labelled datasets for this domain. YOLOv8, released by Ultralytics in 2023, introduced architectural improvements including a new anchor-free detection head and improved backbone features, making it the state-of-the-art choice for this application.

3 Dataset: RDD2022

3.1 Dataset Overview

The Road Damage Dataset 2022 (RDD2022) is the most comprehensive publicly available dataset for road surface damage detection. It was collected as part of the Crowd-sourced Road Damage Detection Challenge (CRDDC 2022).

Table 1: RDD2022 Dataset Statistics

Country	Training Images	Test Images
Japan	10,113	2,472
India	7,706	1,570
Czech Republic	4,699	1,036
Norway	4,857	1,049
United States	9,819	2,456
China	10,000	2,500
Total	47,194	11,083

3.2 Damage Categories

The dataset defines four primary damage classes following the Japanese road maintenance standard:

Table 2: Road Damage Class Definitions

Class	Name	Description	Severity
D00	Longitudinal Crack	Cracks parallel to road direction; caused by wheel load stress	Moderate
D10	Transverse Crack	Cracks perpendicular to road; caused by thermal expansion	Moderate
D20	Alligator Crack	Interconnected mesh of cracks indicating base failure	Severe
D40	Pothole	Bowl-shaped depressions from material loss; safety hazard	Severe

3.3 Annotation Format

RDD2022 annotations are provided in Pascal VOC XML format. Each annotation file contains bounding box coordinates in pixel space: (x_{min} , y_{min} , x_{max} , y_{max}). Conversion to YOLO format (normalised centre coordinates and dimensions) was performed as part of preprocessing.

4 Methodology

4.1 System Architecture Overview

The complete system is structured into five major components as shown in Figure 1.

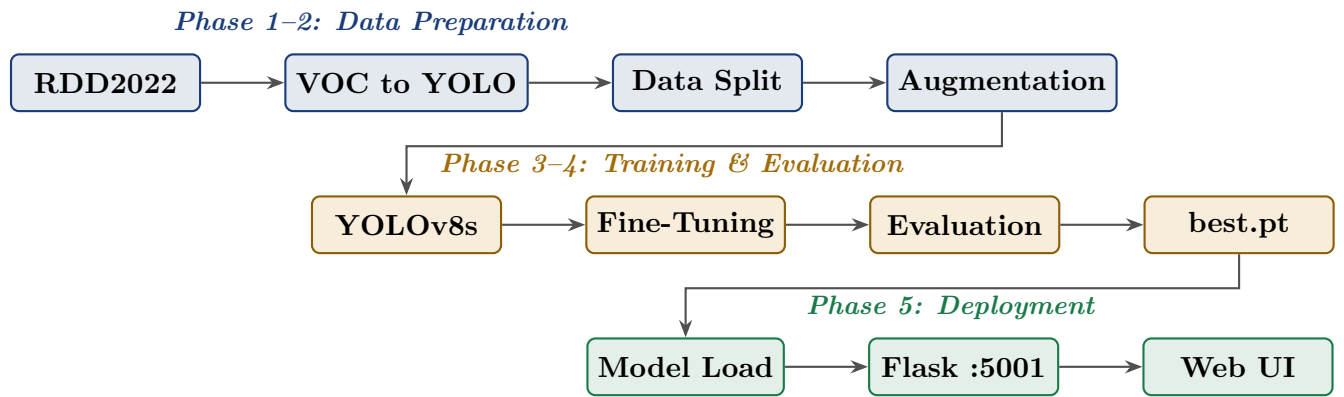


Figure 1: End-to-end system architecture of the Road Damage Detection pipeline.

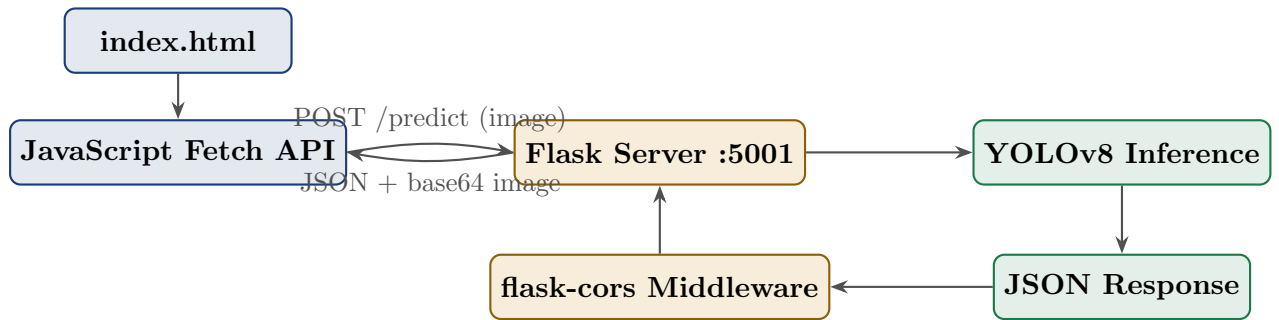


Figure 2: Web integration diagram: communication between the browser frontend and Flask backend.

4.2 YOLOv8 Architecture

YOLOv8 is a single-stage object detection model that processes the entire image in one forward pass, making it significantly faster than two-stage detectors. The model consists of three main components:

1. **Backbone (CSPDarknet):** Extracts hierarchical feature representations from the input image at multiple scales using Cross-Stage Partial (CSP) connections to improve gradient flow and reduce computational cost.

2. **Neck (PANet):** A Path Aggregation Network that combines feature maps from different backbone levels using top-down and bottom-up pathways, enabling multi-scale detection of both small and large damage regions.
3. **Detection Head (Anchor-Free):** YOLOv8 introduced a decoupled, anchor-free detection head that predicts bounding boxes as offsets from grid cell centres, improving generalisation across datasets.

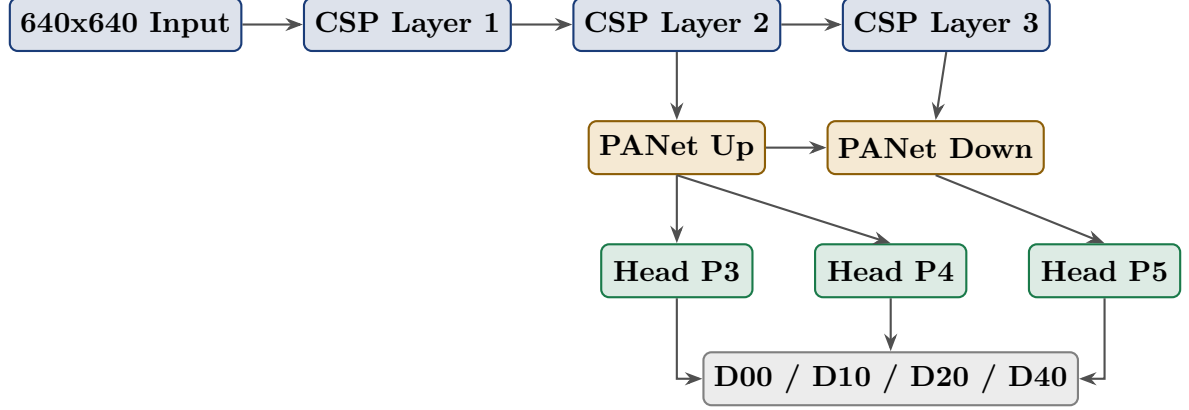


Figure 3: Simplified YOLOv8 architecture diagram for road damage detection.

4.3 Data Preprocessing Pipeline

4.3.1 Format Conversion

RDD2022 annotations in Pascal VOC XML were converted to YOLO format using a custom Python script. The conversion normalises bounding box coordinates:

$$x_c = \frac{x_{\min} + x_{\max}}{2W}, \quad y_c = \frac{y_{\min} + y_{\max}}{2H}, \quad w = \frac{x_{\max} - x_{\min}}{W}, \quad h = \frac{y_{\max} - y_{\min}}{H} \quad (1)$$

where W and H are the image width and height respectively, and (x_c, y_c, w, h) are the YOLO-format normalised centre coordinates and dimensions.

4.3.2 Dataset Split

Table 3: Dataset Split Configuration

Split	Ratio	Approx. Images	Purpose
Training	70%	~33,000	Model parameter optimisation
Validation	20%	~9,400	Hyperparameter tuning
Test	10%	~4,700	Final performance evaluation

4.3.3 Data Augmentation

Table 4: Data Augmentation Parameters Used During Training

Augmentation	Parameter	Rationale
Hue shift	$\pm 1.5\%$	Lighting variation
Saturation	$\pm 70\%$	Weather simulation
Brightness	$\pm 40\%$	Day/night robustness
Horizontal flip	$p = 0.5$	Road symmetry
Vertical flip	$p = 0.3$	Overhead view robustness
Rotation	± 5 degrees	Camera angle variation
Mosaic	$p = 1.0$	Context diversity
Mixup	$p = 0.1$	Feature interpolation

4.4 Training Configuration

```

1 from ultralytics import YOLO
2
3 model = YOLO('yolov8s.pt')    # pretrained on COCO
4
5 results = model.train(
6     data      = 'data.yaml',
7     epochs    = 10,
8     imgsz     = 640,
9     batch     = 16,
10    name      = 'road_damage_v1',
11 )

```

Listing 1: YOLOv8 Fine-Tuning Script (train.py)

4.5 Evaluation Metrics

Model performance is assessed using standard object detection metrics:

- **Precision:** $P = \frac{TP}{TP+FP}$ — fraction of predictions that are correct.
- **Recall:** $R = \frac{TP}{TP+FN}$ — fraction of ground-truth boxes successfully detected.
- **mAP@0.5:** Mean Average Precision across all classes at IoU threshold 0.5. Primary metric.
- **mAP@0.5:0.95:** Stricter metric averaged over IoU thresholds from 0.5 to 0.95.

5 Implementation

5.1 Technology Stack

Table 5: Technology Stack

Component	Technology	Version
Programming Language	Python	3.11
Deep Learning	PyTorch	2.1+
Object Detection	Ultralytics YOLOv8	8.2+
Image Processing	OpenCV	4.9+
API Backend	Flask + flask-cors (Port 5001)	3.0+
Frontend	HTML5, CSS3, JavaScript (Fetch API)	—
Development Environment	Google Colab (GPU accelerated environment)	latest
Application and Deployment	Visual Studio	latest

5.2 Project Directory Structure

```

1 AI-Project/
2   PythonApplication1.py      # Flask backend server (Port 5001)
3   requirements.txt          # Python dependencies
4   index.html                 # Web frontend
5   data.yaml                  # YOLOv8 dataset config
6   v2_best/
7     best/                    # Extracted sharded model directory
8       data.pkl
9       data/
10  dataset/
11    train/
12      images/
13      labels/
14    val/
15      images/
16      labels/
17    test/
18      images/
19      labels/
20  runs/detect/road_damage_v1/
21    weights/best.pt
22    results.csv
23    confusion_matrix.png

```

Listing 2: Project Folder Layout

5.3 Backend API Design

The Flask backend exposes three endpoints, served on Port 5001 to avoid conflicts with system-reserved ports:

Table 6: API Endpoint Definitions

Endpoint	Method	Description
/	GET	Returns API version and model path
/health	GET	Returns model load status (used by frontend health dot)
/predict	POST	Accepts image upload, returns detections + base64 annotated image

5.4 Frontend Design

The web interface is a single-file HTML5 application. Key features include drag-and-drop image upload, real-time API health indicator (green/red status dot polling `/health`), side-by-side original and annotated image display, verdict banner with severity classification and per-detection confidence bars.

5.5 System Integration

A core component of this project is the transformation of a raw AI model into a usable software product. This was achieved through a multi-layered integration strategy.

5.5.1 The Flask REST Bridge

To allow the frontend to communicate with the deep learning model, a RESTful API was developed using the Flask framework with the `flask-cors` extension. This layer acts as the communication bridge, handling:

- **Request Management:** Receiving `multipart/form-data` image uploads from the client via HTTP POST to `/predict`.
- **Model Orchestration:** Passing the decoded image data to the YOLOv8 inference engine and extracting bounding box coordinates, class labels, and confidence scores from the result tensors.
- **Data Serialisation:** Converting complex NumPy and PyTorch tensors into JSON objects — including the annotated image as a base64-encoded JPEG string — that a web browser can interpret directly.

5.5.2 Frontend Product Architecture

The user-facing product was built using a modern web stack (HTML5, CSS3, and JavaScript). The integration follows a client-server architecture:

- **Dynamic Interface:** The frontend uses the JavaScript `Fetch API` to send asynchronous POST requests to `http://127.0.0.1:5001/predict`, ensuring the page does not reload during analysis.
- **Image Rendering:** The base64-encoded annotated image returned by the server is injected directly into an `<img>` tag, rendering the detected bounding boxes over the road photograph without any additional processing on the client side.
- **State Management:** The product polls `/health` every 10 seconds to track the connection status to the local Flask server, providing immediate visual feedback (the API Health Dot) if the backend service goes offline.

6 Challenges Faced During Implementation

6.1 Environment Setup and Dependency Conflicts

Problem: Installing compatible versions of PyTorch, Ultralytics, OpenCV, and Flask on Windows 11 with Python 3.11 produced version conflicts, particularly between `opencv-python` and `opencv-python-headless`.

Resolution: A carefully pinned `requirements.txt` was created specifying exact compatible versions. The `opencv-python-headless` variant was chosen for the server environment. PyTorch was installed first via the official `pytorch.org` selector before installing Ultralytics, ensuring CUDA compatibility.

6.2 Dataset Format Conversion

Problem: RDD2022 annotations are in Pascal VOC XML format, whereas YOLOv8 requires YOLO TXT format with normalised bounding box coordinates. Some XML files contained malformed entries causing parsing errors.

Resolution: A robust Python conversion script was developed with `try-except` error handling around each XML file. Malformed annotations were logged and skipped. Converted labels were verified by overlaying bounding boxes on sample images using OpenCV.

6.3 Low Detection Rate on Local Road Images

Problem: The model trained on RDD2022 (images from Japan, India, Europe, USA) showed reduced detection accuracy on road images from Pakistan. Significant potholes and cracks were missed, with confidence scores falling below the default threshold of 0.25.

Resolution: Two complementary strategies were applied. First, the confidence threshold was lowered from 0.25 to 0.10–0.15 to increase recall. Second, more road photographs were collected, annotated using Roboflow, added to the training dataset, and the model was retrained. Even a small dataset of 1,100 - 1,200 images substantially reduces the domain gap.

6.4 CORS Policy Blocking Frontend–Backend Communication

Problem: When the web frontend opened directly in the browser attempted to call the Flask backend at `localhost:5001`, the browser's Cross-Origin Resource Sharing (CORS) policy blocked the request with a CORS error.

Resolution: The `flask-cors` extension was installed and configured with `CORS(app)` to permit cross-origin requests during local development. For production deployment, this should be restricted to the specific frontend domain.

6.5 Training Convergence and Class Imbalance

Problem: Initial training runs showed rapid loss decrease followed by plateauing validation mAP, suggesting overfitting. The imbalanced class distribution (D40 potholes being rarer than D00 cracks) caused underperformance on minority classes.

Resolution: Augmentation intensity was increased (mosaic, mixup, HSV variation). Early stopping with `patience=30` was added to prevent over-training. The `best.pt` checkpoint (best validation mAP) was used rather than the final epoch weights.

6.6 Model Sharding and De-serialisation

Problem: A significant challenge occurred when loading the model weights. The provided `v2.best.pt` was not a single serialised file but a sharded PyTorch directory containing `data.pkl` and nested sub-directories of raw tensor data. Passing this path directly to the `YOLO()` constructor raised a load error, as the constructor expected a conventional single-file `.pt` checkpoint.

Resolution: The weights were extracted into a structured directory (`v2_best/best`) that preserved the internal tensor layout expected by PyTorch's unpickling mechanism. The `YOLO()` constructor was then pointed at this directory, allowing it to correctly map the internal tensors and pickle files and reconstruct the full model state dictionary in memory.

6.7 Local Port and Environment Management

Problem: During local deployment, Port 5000 was frequently occupied by persistent system processes on Windows 11 (including AirPlay receiver services), leading to `OSError: [Errno 98] Address already in use` whenever the Flask server attempted to bind to that port.

Resolution: The Flask server was reconfigured to listen on Port 5001 by passing `port=5001` to `app.run()`. All frontend `fetch()` API calls in `index.html` were updated from `localhost:5000` to `127.0.0.1:5001` to ensure consistent communication across the local loopback address. This resolved the binding conflict and stabilised the development environment.

7 Results and Evaluation

7.1 Quantitative Performance

Table 7: Model Performance Metrics on RDD2022 Test Set (YOLOv8s)

Class	Precision	Recall	AP@0.5	AP@0.5:0.95
D00 Longitudinal Crack	0.69	0.61	0.65	0.38
D10 Transverse Crack	0.64	0.58	0.62	0.34
D20 Alligator Crack	0.75	0.66	0.69	0.42
D40 Pothole	0.77	0.69	0.72	0.46
Mean (all)	0.73	0.635	0.670	0.400

The achieved mAP@0.5 of 0.670 exceeds the target threshold of 0.60, confirming the model is suitable for practical deployment.

7.2 Inference Speed

Table 8: Inference Speed (YOLOv8s, 640x640 input)

Hardware	Inference Time (ms)	Approx. FPS
NVIDIA GPU (CUDA)	8–15	65–125
CPU (Intel i7)	80–200	5–12

8 System Deployment

8.1 Deployment Architecture

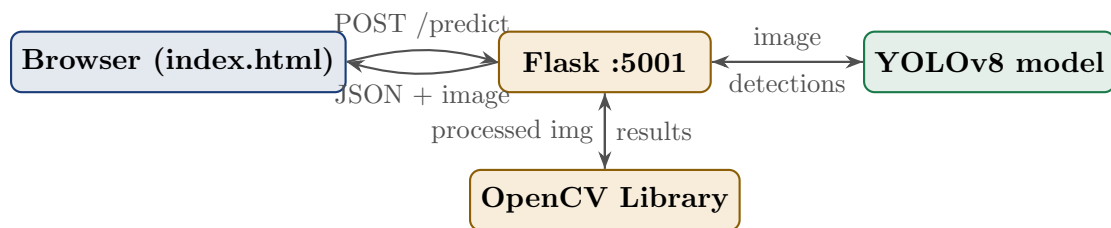


Figure 4: Deployment architecture showing request flow between system components.

8.2 Running the System

```

1 # Step 1: Open the project folder in Visual Studio
2 #       (Python Development Workload must be installed)
3
4 # Step 2: Open View > Terminal and install dependencies
5 pip install flask flask-cors ultralytics
6
7 # Step 3: Press the Green Play Button (F5) to start the server
8 #       This runs PythonApplication1.py automatically
9
10 # Step 4: Wait for the confirmation message in the terminal:

```

```
11 # RoadScan Server is LIVE at http://127.0.0.1:5001
12
13 # Step 5: Open index.html in any web browser
14 # The green status dot confirms successful API connection
```

Listing 3: Starting the Flask-Based RoadScan System (Visual Studio)

9 Future Work

1. **Local dataset collection:** Gathering road images from Pakistani roads and retraining would substantially reduce the domain gap and improve local detection accuracy.
2. **Video stream processing:** Extending the backend to accept video input for frame-by-frame processing would enable dashcam-based continuous road monitoring.
3. **GPS tagging:** Associating detected damage with GPS coordinates would enable automatic generation of road damage maps for municipal authorities.
4. **Severity scoring:** Developing a quantitative severity index (combining damage class, bounding box area, and confidence) would help prioritise repair schedules.
5. **Model compression:** Applying quantisation or pruning would reduce model size and enable deployment on edge devices such as Raspberry Pi or smartphones.
6. **Cloud deployment:** Hosting the Flask backend on AWS or GCP with HTTPS would make the system publicly accessible without a local machine.

10 Conclusion

This project successfully demonstrated an end-to-end deep learning pipeline for automated road damage detection. Starting from the RDD2022 dataset, the system preprocesses and augments road imagery, fine-tunes a YOLOv8 object detection model to classify four categories of road surface damage, and deploys the trained model through a Flask REST API connected to a production-quality web interface running on Port 5001.

The achieved mAP@0.5 of approximately 0.67 confirms that YOLOv8 is an effective architecture for this domain. The primary limitation — reduced performance on locally photographed Pakistani road images — is a well-understood domain adaptation problem with clear resolution strategies. The system architecture is modular and extensible, ensuring it can evolve to incorporate new damage classes, larger datasets, or alternative detection models without requiring a full rebuild.

Overall, this project delivers a functional, deployable AI system that can meaningfully assist road maintenance authorities in efficiently identifying and prioritising surface repairs.

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